THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2310	Highest powers of $2 \times 3 \times 5$, $2 \times 3 \times 7$, $2 \times 3 \times 11$
2	**(a)** 60	LCM of 1 to 6
3	**(b)** 9	HCF of the differences from the remainder
4	**(b)** 3	$12 = 2^2 \times 3$; multiply by the primes that occur to an odd power
5	**(b)** $\frac{\sqrt{12}}{\sqrt{3}}$	$\sqrt{12}/\sqrt{3} = 2$
6	**(a)** $2^3 \times 5^2 \times 3$	Zeros at the end = min(power of 2, power of 5)
7	**(d)** 144	Highest powers of $2^2 \times 3^2$, $2^4 \times 3$, $2^3 \times 3^2$
8	**(d)** 420	LCM of 1 to 7
9	**(a)** Both A and R are true and R is the correct explanation of A.	HCF x LCM = 6 x 36 = 216 = 12 x 18, and R is the general result used to find the LCM.
10	**(a)** Both A and R are true and R is the correct explanation of A.	5 is rational and $2\sqrt{3}$ is irrational, so the difference is irrational; R explains it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Numbers are $12 \times 3 = 36$ and $12 \times 5 = 60$ (1); LCM $= 180$ (1).	1+1
12	Each remainder is short of the divisor by 5 (1); LCM$(12, 18) = 36$, number $= 36 - 5 = 31$ (1).	1+1
13	If $3\sqrt{7} = r$ were rational, $\sqrt{7} = r/3$ would be rational (1) - a contradiction (1).	1+1
14	Numbers are $15 \times 4 = 60$ and $15 \times 7 = 105$ (1); LCM $= 420$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM$(18, 24, 36) = 72$ min (2); rounds $4, 3, 2$ (1).	2+1
16	Required number $=$ HCF$(1250, 9376)$ (1); HCF$(1250, 9376) = 2$ (2).	1+2
17	Take 7 common: $7(5 \times 6 + 1)$ (2); $5 \times 6 + 1 = 31 > 1$, so there are two factors greater than 1 (1); composite.	2+1
	OR: LCM$(20, 28, 35) = 140$ min (2); rounds $7, 5, 4$ (1).	2+1
18	Required number $=$ HCF$(1152, 1392)$ (1); HCF$(1152, 1392) = 48$ (2).	1+2
	OR: Take 8 common: $8(6 \times 7 + 1)$ (2); $6 \times 7 + 1 = 43 > 1$, so there are two factors greater than 1 (1); composite.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $144 = 2^4 \times 3^2$, $180 = 2^2 \times 3^2 \times 5$ (1). (ii) HCF = $2^2 \times 3^2 = 36$ chairs per row (1). (iii) Rows: $144/36 + 180/36 = 4 + 5 = 9$ (1). (iv) $216 = 2^3 \times 3^3$; HCF(144, 180, 216) = $2^2 \times 3^2 = 36$ (1); rows $= 4 + 5 + 6 = 15$ (1).	1+1+1+2
20	Books per stack $=$ HCF$(144, 192, 288) = 48$ (3); stacks $= \frac{624}{48} = 13$ (2).	3+2
	OR: If $1/\sqrt{3} = r$ were rational then $\sqrt{3} = 1/r$ would be rational - contradiction (3); $\frac{5}{\sqrt{3}} = 5 \times \frac{1}{\sqrt{3}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2

Mistakes to avoid:

- Using HCF × LCM = product for **three** numbers; the result holds only for two numbers.
- Taking the *greatest* power for HCF or the *smallest* power for LCM; HCF uses smallest common powers, LCM uses greatest powers.
- Checking the denominator before reducing the fraction to lowest terms when testing for a terminating decimal.
- Assuming the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{2} = 2$.
- In irrationality proofs, forgetting to state that a and b are co-prime at the start, which is what gives the contradiction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{\sqrt{75}}{\sqrt{3}}$	$\sqrt{75}/\sqrt{3} = 5$
2	**(c)** $2^4 \times 5^3 \times 3$	Zeros at the end = min(power of 2, power of 5)
3	**(c)** 900	Highest powers of $3^2 \times 5$, $2^2 \times 3 \times 5$, 3×5^2
4	**(a)** 840	LCM of 1 to 8
5	**(a)** 20	HCF of the differences from the remainder
6	**(a)** 5	$45 = 3^2 \times 5$; multiply by the primes that occur to an odd power
7	**(c)** $\frac{\sqrt{108}}{\sqrt{3}}$	$\sqrt{108}/\sqrt{3} = 6$
8	**(d)** $2^5 \times 5^4 \times 3$	Zeros at the end = min(power of 2, power of 5)
9	**(c)** A is true but R is false.	$\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$ is rational, so A is true; R is false (this is a counter-example).
10	**(a)** Both A and R are true and R is the correct explanation of A.	A rational number whose denominator (in lowest terms) is of the form $2^m 5^n$ has a terminating decimal expansion.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Each remainder is short of the divisor by 4 (1); LCM$(15, 20) = 60$, number $= 60 - 4 = 56$ (1).	1+1
12	If $5\sqrt{2} = r$ were rational, $\sqrt{2} = r/5$ would be rational (1) - a contradiction (1).	1+1
13	Numbers are $9 \times 5 = 45$ and $9 \times 8 = 72$ (1); LCM $= 360$ (1).	1+1
14	Each remainder is short of the divisor by 6 (1); LCM$(10, 14) = 70$, number $= 70 - 6 = 64$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM$(15, 40, 60) = 120$ min (2); rounds $8, 3, 2$ (1).	2+1
16	Required number $=$ HCF$(2010, 2514)$ (1); HCF$(2010, 2514) = 6$ (2).	1+2
17	Take 9 common: $9(7 \times 8 + 1)$ (2); $7 \times 8 + 1 = 57 > 1$, so there are two factors greater than 1 (1); composite.	2+1
	OR: LCM$(16, 24, 30) = 240$ min (2); rounds $15, 10, 8$ (1).	2+1
18	Required number $=$ HCF$(1500, 1920)$ (1); HCF$(1500, 1920) = 60$ (2).	1+2
	OR: Take 10 common: $10(8 \times 9 + 1)$ (2); $8 \times 9 + 1 = 73 > 1$, so there are two factors greater than 1 (1); composite.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $36 = 2^2 \times 3^2$, $48 = 2^4 \times 3$ (1). (ii) LCM = $2^4 \times 3^2 = 144$ s (1). (iii) Aman: $144/36 = 4$ rounds; Bhavna: $144/48 = 3$ rounds (1). (iv) $60 = 2^2 \times 3 \times 5$; LCM(36, 48, 60) = $2^4 \times 3^2 \times 5 = 720$ s (1); Chetan: $720/60 = 12$ rounds (1).	1+1+1+2
20	Books per stack $=$ HCF$(168, 252, 420) = 84$ (3); stacks $= \frac{840}{84} = 10$ (2).	3+2
	OR: If $1/\sqrt{7} = r$ were rational then $\sqrt{7} = 1/r$ would be rational - contradiction (3); $\frac{4}{\sqrt{7}} = 4 \times \frac{1}{\sqrt{7}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2

Mistakes to avoid:

- Using HCF × LCM = product for **three** numbers; the result holds only for two numbers.
- Taking the *greatest* power for HCF or the *smallest* power for LCM; HCF uses smallest common powers, LCM uses greatest powers.
- Checking the denominator before reducing the fraction to lowest terms when testing for a terminating decimal.
- Assuming the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{2} = 2$.
- In irrationality proofs, forgetting to state that a and b are co-prime at the start, which is what gives the contradiction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 28	HCF of the differences from the remainder
2	**(c)** 7	$28 = 2^2 \times 7$; multiply by the primes that occur to an odd power
3	**(d)** $\frac{\sqrt{147}}{\sqrt{3}}$	$\sqrt{147}/\sqrt{3} = 7$
4	**(d)** $2^6 \times 5^5 \times 3$	Zeros at the end = min(power of 2, power of 5)
5	**(a)** 252	Highest powers of $2^2 \times 7$, $2 \times 3 \times 7$, $3^2 \times 7$
6	**(a)** 2520	LCM of 1 to 10
7	**(c)** 14	HCF of the differences from the remainder
8	**(d)** 10	$90 = 2 \times 3^2 \times 5$; multiply by the primes that occur to an odd power
9	**(a)** Both A and R are true and R is the correct explanation of A.	LCM = (product)/HCF = product when HCF = 1.
10	**(d)** A is false but R is true.	$6^n = 2^n 3^n$ has no factor 5, so it never ends in 0; A is false, R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	If $4\sqrt{11} = r$ were rational, $\sqrt{11} = r/4$ would be rational (1) - a contradiction (1).	1+1
12	Numbers are $18 \times 2 = 36$ and $18 \times 7 = 126$ (1); LCM $= 252$ (1).	1+1
13	Each remainder is short of the divisor by 7 (1); LCM$(16, 24) = 48$, number $= 48 - 7 = 41$ (1).	1+1
14	If $6\sqrt{3} = r$ were rational, $\sqrt{3} = r/6$ would be rational (1) - a contradiction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM$(22, 33, 44) = 132$ min (2); rounds $6, 4, 3$ (1).	2+1
16	Required number $=$ HCF$(1812, 2412)$ (1); HCF$(1812, 2412) = 12$ (2).	1+2
17	Take 11 common: $11(9 \times 10 + 1)$ (2); $9 \times 10 + 1 = 91 > 1$, so there are two factors greater than 1 (1); composite.	2+1
	OR: LCM$(14, 35, 49) = 490$ min (2); rounds $35, 14, 10$ (1).	2+1
18	Required number $=$ HCF$(3000, 3420)$ (1); HCF$(3000, 3420) = 60$ (2).	1+2
	OR: Take 12 common: $12(10 \times 11 + 1)$ (2); $10 \times 11 + 1 = 111 > 1$, so there are two factors greater than 1 (1); composite.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $125 = 5^3$ (1). (ii) In lowest terms the denominator must be of the form $2^m 5^n$ (1). (iii) Only $\frac{13}{125}$ terminates; $18 = 2 \times 3^2$ and $343 = 7^3$ contain primes other than 2 and 5 (1). (iv) $\frac{13}{125} = \frac{13 \times 8}{1000} = 0.104$ (1); $\frac{17}{18}$ does not terminate because its denominator has the prime factor 3 (1).	1+1+1+2
20	Books per stack $=$ HCF$(108, 162, 270) = 54$ (3); stacks $= \frac{540}{54} = 10$ (2).	3+2
	OR: If $1/\sqrt{13} = r$ were rational then $\sqrt{13} = 1/r$ would be rational - contradiction (3); $\frac{6}{\sqrt{13}} = 6 \times \frac{1}{\sqrt{13}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2

Mistakes to avoid:

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